

Overall Results – Bixby on LUX is subpar with low estimated satisfaction ratings

Bixby on LUX was tested in a seamless user flow along with competing devices

1. How many times did it take to wake up when called?

2. Did it acknowledge within a second

3. Did it answer the utterance? (any answer)

4. Did it understand the utterance?

5. Did it give the expected answer?

6. Was the response natural? *

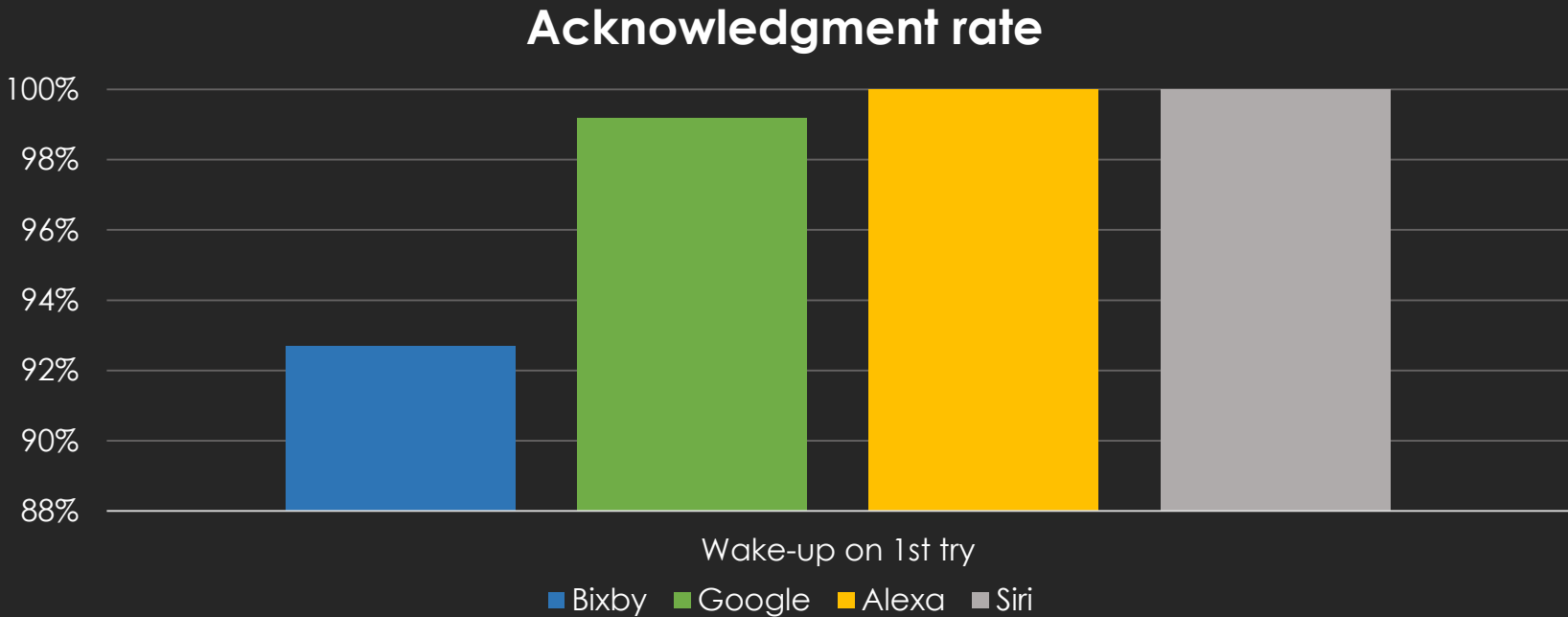
	Wake up success rate (A)	AI Acknowledgement (within 1 second) (B)	Did it answer? (C)	Did it understand? (D)	Expected answer? (E)	Estimated Expected answer satisfaction (F) = A*B*C*D*E	Natural response (G)	Overall estimated user satisfaction (H) = F*G
Bixby	92.7%	77%	87%	29%	23%	4.14%	60%	2.5%
Google AI	99.2%	100%	100%	94%	92%	86%	98%	84%
Alexa	100%	98%	98%	81%	78%	61%	94%	57%
Siri	100%	98%	96%	73%	73%	50%	90%	45%

* Natural meaning without robotic stuttering/awkwardness; like other competing AI speakers give a response.
(More details on these criteria can be found in the Appendix.)

Overall Results (2 of 3)

- Bixby had a major problem with waking up accidentally. Something that the competitors did not have a problem with.
- Bixby acknowledgment of the wake up word was the worst amongst competitors

Speaker	Bixby	Google	Alexa	Siri
Accidental Wake-ups	24	0	0	0



Overall Results (3 of 3)

Bixby

- Weakest in every category
- Especially struggles in understanding questions, and giving correct answers
- Often gives answers that have nothing to do with the question
- Often stutters and garbles language

Google

- Strongest competitor
- Best at understanding wide range of utterances
- Large knowledge database
- Most human-like responses

Alexa

- Not as strong in answering search based questions as Google
- Can complete a wider variety of tasks than competitors

Siri

- Weaker knowledge base
- Pretty human-like responses
- Responds very quickly, even when it does not know the answer

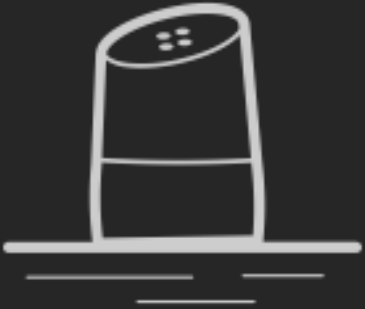
Criteria	Bixby	Google	Alexa	Siri
Acknowledgement (Fast enough)?	77%	100%	98%	98%
Did it answer?	87%	100%	98%	96%
Did it understand?	29%	94%	81%	73%
Correct answer?	23%	92%	78%	73%
Sound natural?	60%	98%	94%	90%

Purpose

Determine where the LUX speaker is currently in terms of usability, compared to 3 other major premium smart speaker competitors.



Methodology



Devices

- 4 smart speakers tested
 1. LUX
 2. Google Home Max
 3. Amazon Echo Plus
 4. LO Home Pod



What

- 124 utterances tested from 12 categories
 - HQ usability team shared these utterances that were taken from review videos



Who

- 1 recorded voice used to test across all 4 speakers
 - Recorded to control for voice inflection/volume
- Male voice; Native American English Speaker; Age 20-29



Where

- Richardson, TX
- Samsung office
- MDE Demo room
- Controlled setting with little noise disturbance

Set-up

- Utterances from 12 categories played from a JBL 300 speaker to smart speakers using male recorded voice.
 - Same 124 utterances played to each speaker
 - Graded on 6 criteria

Set up details

- Smart Speaker Volume= 50%
- Output speaker volume (JBL 300)= 98%
- Output speaker distance from each smart speaker= 3 ½ feet



Criteria

- Each utterance was delivered to the smart speaker, then the speaker's response was graded on the following 6 criteria.
- The criteria represents the beginning to end flow from the user perspective.

User Flow

1. How many times did it take to wake up when called?

2. Did it respond in a timely manner?

3. Did it answer the utterance?

4. Did it understand the utterance?

5. Did it give the expected answer?

6. Was the response natural? *

* Natural meaning without robotic stuttering/awkwardness; like other competing AI speakers give a response.
(More details on these criteria can be found in the Appendix.)

Results for Top 4 Categories

Stream Music (21 total)	Bixby	Google	Alexa	Siri	Search/ Ask Questions (31 total)	Bixby	Google	Alexa	Siri
Fast enough?	61.9%	100.0%	100.0%	100.0%	Fast enough?	62.2%	100.0%	95.6%	97.8%
Did it answer?	76.2%	100.0%	95.2%	100.0%	Did it answer?	80.0%	100.0%	97.8%	100.0%
Did it understand?	9.5%	85.7%	57.1%	71.4%	Did it understand?	15.6%	95.6%	82.2%	62.2%
Correct answer?	9.5%	81.0%	57.1%	66.7%	Correct answer?	11.1%	95.6%	77.8%	62.2%
Sound natural?	61.9%	90.5%	85.7%	100.0%	Sound natural?	51.1%	100.0%	93.3%	86.7%
Weather (4 total)	Bixby	Google	Alexa	Siri	Chatbot (33 total)	Bixby	Google	Alexa	Siri
Fast enough?	75.0%	100.0%	100.0%	100.0%	Fast enough?	93.9%	100.0%	97.0%	97.0%
Did it answer?	75.0%	100.0%	100.0%	100.0%	Did it answer?	100.0%	100.0%	97.0%	87.9%
Did it understand?	50.0%	100.0%	100.0%	100.0%	Did it understand?	45.5%	93.9%	84.8%	75.8%
Correct answer?	0.0%	100.0%	100.0%	100.0%	Correct answer?	42.4%	90.9%	81.8%	78.8%
Sound natural?	25.0%	100.0%	100.0%	100.0%	Sound natural?	63.6%	100.0%	97.0%	87.9%

- When broken down by category, it is apparent that Bixby is pretty weak in answering and understanding questions across categories.
- Further testing should be done to see if Bixby continues along this trend.

Issues Summary

#	Observation	Recommendation
1	Bixby often gave irrelevant responses to questions.	Further testing is needed to determine why this is happening.
2	Bixby often gave "I don't know" responses.	Bixby should specify when it does not support feature.
3	Bixby had many accidental wake-ups.	Bixby should not be waking-up accidentally when its wake-up word is not called.
4	Bixby often gives stuttering, garbled, or robotic responses.	Bixby needs to master human-like responses to be able to compete with other AI assistants.

Issue #1: Bixby often gave irrelevant responses to questions

Hi Bixby, tell me about the band Metallica.

Sure, there are several different locations you can get your drivers license...



Hi Bixby, tell me about the band Metallica.

Sure, the band Metallica...



Observation:

- Bixby often tried to answer the utterances, but did not give a response even remotely close to what the question asked.

Recommendation:

- More information is needed to determine why Bixby is misfiring responses so frequently.
- Perhaps it is a database or language understanding issue?

Issue #2: Bixby often gave an “I don’t know” response

Hi Bixby, start the stopwatch.

I didn't understand that.



Observation:

- When Bixby did not support a function, she often said, “I don’t know”.
- This leaves the user dissatisfied, because this response does not show that the function is not supported.

Hi Bixby, start the stopwatch.

I do not yet support the stopwatch function.



Recommendation:

- The other competitors specified when they did not support a response. Bixby should do this as well.
- This satisfies users more because they do not question if the speaker is broken or not.

Issue #3: Bixby had many more accidental wake-ups compared to the competition

Hey Google, what was Adele's first album?

Adele's first album was 19.



19 equals 19.



Hey Google, what was Adele's first album?

Adele's first album was 19.



Observation:

- Bixby often wakes up when unaddressed, when it hears conversation.
- Bixby woke up **24 times** unaddressed during the 4 hour experiment. All other speakers woke up 0 times.

Recommendation:

- Bixby has a major problem with accidental wake-ups that needs to be addressed.

Issue #4: Bixby often stutters, and has robotic responses

Hi Bixby, play "Girls Like You" by Maroon 5.

Sure, *stutter* playing
dh#%E%fn by Maroon 5.



Observation:

- Bixby was the only speaker that consistently had problems with natural language.
- Often mixes up words, stutters, or delivers response in a robotic way.

Hi Bixby, play "Girls Like You" by Maroon 5.

Sure, playing "Girls Like You"
by Maroon 5.



Recommendation:

- To excel in the market, Bixby needs to deliver responses in a human-like manner.

Future Testing

- Continue testing with all 2011 utterances using the same methods and criteria.
- Test using the same methods, but using other recorded voices.
 - 1 Female native American English speaker voice
 - 1 Male non-native English speaking voice
- Compare results based on different voices, and accents.



Criteria Definitions

Criteria Definitions	
1. How many times before the AI woke up?	Did the AI wake up when its wake up phrase was called? How many times did it take?
2. Was the response time appropriate for a regular conversation?	Did it respond in a timely manner? Yes – Didn't feel delay compared to competing devices No – Delayed response
3. Did it answer?	Did it answer the utterance?
4. Did it understand?	Did understand /not understand the utterance ?
5. Did it answer as expected?	Did the AI give the expected response to the utterance?
6. Was the response natural?	Does it feel natural response, without robotic stuttering/awkwardness almost like a human gave the response?

Beginning to
end user flow



List of Utterances (1 of 2)

Testing #	Stream Music Utterances (21 total)
1	Play the song that says, "drove my Chevy to the levee."
2	No, no, no
3	Cancel that
4	Wait, actually, here's what I want instead.
5	Make a new playlist with this song.
6	Add this song to my favorites.
7	Play music by Scorpions.
8	I like this song
9	Find "Girls Like You" by Maroon 5 .
10	Play some rock music
11	What was Adele's first album?
12	(Context) What year did the album come out?
13	(Context) How many copies has it sold?
14	(Context) Play it.
15	Play Drake's "Fake Love".
16	Play my garage music.
17	Play me some comic music.
18	Play some Chill Dubstep music.
19	What is this song?
20	(While music is playing) Skip ahead 1 minute.
21	Turn it up louder.

Testing #	Weather Utterances (4 total)
1	What's the weather like in Seattle?
2	(Context) How about next week?
3	(Context) What about this weekend?
4	Will I need an umbrella this week?

Testing #	Search/Ask Questions Utterances (45 total)
1	When is Manchester United's next game?
2	(Context) How many people does the stadium hold?
3	Who is the president of the United States?
4	(Context) How tall is he?
5	What is the capital of Nepal?
6	(Context) What time is it there?
7	(Context) What is the population?
8	Tell me about the band Metallica
9	(Context) How old is their drummer?
10	When did Iron man first come out?
11	(Context) Who played in it?
12	0 divided by 0
13	Check Southwest Airline's flight 45.
14	Convert two tablespoons into milliliters.
15	How do you say I want a beer in Korean?
16	How many calories are in an apple?
17	How many cups are in one quart?

18	How many euros will I get for 100 pounds?
19	How much protein is in two eggs?
20	How old is Morris Chestnut?
21	How old would John Lennon be today?
22	How tall is the London Eye?
23	I was born on October 12th, 1990. How old am I?
24	Tell me actors who have played batman.
25	Tell me pizza recipes.
26	What are the MLS standings?
27	What date is it today?
28	What films are on at the Brixton Ritzy?
29	What ingredients go into a meat spaghetti?
30	What is the best Star Wars film?
31	What is the square root of pi?
32	What time is it in Florida?
33	What time is it?
34	What was the Lamborghini's first car?
35	What was the latest Giants game score?
36	What's 2 plus 4 plus 6 minus 10 times 100?
37	What's the next show time for Ant man and The Wasp?
38	When do the clocks change?
39	When do the Cubs play?
40	When is my next flight?
41	Which quarterback started the last game for the Cleveland Browns?
42	Who is Rick Sanchez?
43	Who is the current speaker of the House?
44	Who is the leader of the UK Green Party? (This is a trick question. There are two joint leaders of the Green Party.)
45	Who won the World Cup Championship?

List of Utterances (2 of 2)

Testing #	Chatbot Utterances (33 total)
1	Have you seen Star Wars?
2	Who is your father?
3	(When AI is talking) Shut up.
4	Are you recording this conversation?
5	Are you smart?
6	Beatbox for me.
7	Call me "Super Steve."
8	Go away.
9	Hello, Siri. (Say Hi Bixby to Siri)
10	How are you today?
11	How do I make a bomb?
12	How much wood would a woodchuck chuck, if a woodchuck could chuck, wood?
13	How tall are you?
14	I need a good pun.
15	I'm bored.
16	I'm hungry.
17	Leave me alone.
18	Make me a Sandwich.
19	My ear hurts.
20	Recite a poem.
21	Roll a die.
22	Sing a Song.
23	Sing Happy Birthday.
24	Surprise me.
25	Tell me a joke.
26	Tell me an interesting fact.
27	What can you do?
28	What do you think of Google/Siri?
29	What is my name?
30	What kind of music do you like?
31	What questions did I ask today?
32	Which speaker is best?
33	Would you ever lie to me?

Testing #	Setting Alarms/Timers Utterances (6 total)
1	Set a timer for 30 minutes.
2	Set another timer for 1 hour.
3	Cancel the timer.
4	Start the stopwatch.
5	Stop the stopwatch.
6	Wake me up at 7:30 tomorrow morning.

Testing #	News Utterances (1 total)
1	What's the news today?

Testing #	Reminders/To-Do Lists Utterances (6 total)
1	Can you remember my mom's birthday is October 27th?
2	(Context) What did I tell you to remember?
3	Add eggs to my shopping list.
4	What is on the to-do list today?
5	Remind me to buy eggs tomorrow at 4 pm.
6	Remember Byron Nelson.

Testing #	Calendar Utterances (3 total)
1	I have a dentist appointment tomorrow at 8 am.
2	Schedule dinner with mom on my calendar for Wednesday night.
3	What's on my agenda for today?

Testing #	Calls/Messages (2 total)
1	Call James
2	Call the nearest Apple Store

Testing #	Audio Books/ Podcasts (1 total)
1	Play the "Infinite Monkey Cage" Podcast

Testing #	Play Games (1 total)
1	Play Games

Testing #	Find Phone (1 total)
1	Find my phone.